



DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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PRELIMINARY EXAMINATION 2023 SECONDARY 4 NORMAL ACADEMIC

MATHEMATICS SYLLABUS A

4045/01

Paper 1

27 July 2023

2 hours

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Calculator Model

Answer **all** questions.

If working is needed for any question it must be shown with the answer.

Omission of essential working will result in loss of marks.

The total of the marks for this paper is 70.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator value or 3.142, unless the question requires the answer in terms of π .

Total Marks

Mathematical Formulae*Compound interest*

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

$$\text{Area of a triangle } ABC = \frac{1}{2} ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f}$$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f} \right)^2}$$

Answer **all** the questions.

- 1 By writing each value correct to 1 significant figure, estimate the value of $\frac{19.86 \times 0.0145}{\sqrt{137}}$.
Show your working.

Answer [2]

- 2 Find the smallest prime number x satisfying $-3x \leq 24$.

Answer [2]

3 $4^{\frac{5}{2}} \div 8^3 = 2^n$

Find the value of n .

Answer $n = \dots\dots\dots$ [2]

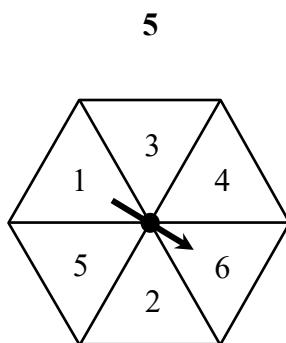
- 4 A is the point $(2, 1)$ and B is the point (p, q) , where p and q are both positive integers more than 10 but not more than 20.

The gradient of the line AB is $\frac{4}{5}$.

Find the coordinates of B .

Answer $(\dots\dots\dots, \dots\dots\dots)$ [2]

5



Kelly spins this fair spinner.

Calculate the probability that

- (a) it stops on a number less than 3,

Answer [1]

- (b) it stops on a factor of 6.

Answer [1]

- 6 Amelia records the number of minutes she spends on social media each day for 30 days.

The stem-and-leaf diagram shows her results.

1		8	9				
2		0	0	3	5	6	
3		1	2	2	2	4	7
4		5	5	6	6	7	8
5		3	3	4	9	9	9
6		0	1	7			
7		2					

Key 1 | 8 means 18

Find

- (a) the number of days in which Amelia spends at most an hour on social media,

Answer [1]

- (b) the median.

Answer minutes [1]

7 The floor plan of an office is drawn to a scale of 1 : 50 .

(a) The length of the office on the plan is 24 cm.

Find the actual length, in metres, of the office.

Answer m [1]

(b) The area of a storeroom is 10.2 m^2 .

Work out the area, in cm^2 , of the storeroom on the plan.

Answer cm^2 [2]

8 $\sin x^\circ = 0.65$

Find two possible values of x .

Answer $x = \dots\dots\dots$ or $\dots\dots\dots$ [2]

- 9 (a) Find the equation of the line of symmetry of $y = -(x-1)(x+6)$.

Answer [2]

- (b) Find the coordinates of the minimum or maximum point of $y = -(x-1)(x+6)$.

Answer (.....,) [1]

- 10 A car travels on a journey of 28 km at an average speed of 48 km/h.

- (a) How many minutes does the journey take?

Answer minutes [2]

- (b) Write 48 km/h as a speed in m/s.

Answer m/s [1]

11 Belle owns a furniture shop.

- (a)** She sold a sofa for \$1275 after a discount of 15%.

Calculate the price of the sofa before the discount.

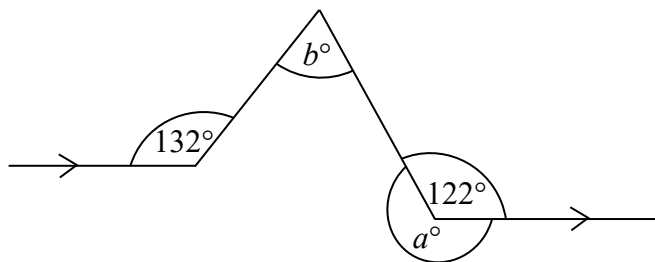
Answer \$ [2]

- (b)** Belle paid \$70 for a chair.
She made a profit when she sold it for \$98.

Find the profit as a percentage of the cost price.

Answer % [2]

12



- (a) Find a , giving a reason for your answer.

$a = \dots\dots\dots$ because $\dots\dots\dots$ [2]

- (b) Calculate b .

Answer $b = \dots\dots\dots$ [2]

13 $x^2 + 12x + 25 = (x + h)^2 + k$

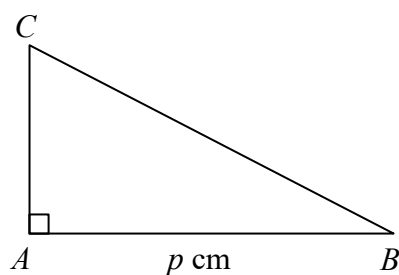
- (a) Find h and k .

Answer $h = \dots\dots\dots, k = \dots\dots\dots$ [2]

- (b) Hence solve $x^2 + 12x + 25 = 0$, giving your answers correct to 2 decimal places.

Answer $x = \dots\dots\dots$ or $\dots\dots\dots$ [2]

14



ABC is a right-angled triangle.

The length of AB is p cm.

AC is 14 cm shorter than AB .

BC is 2 cm longer than AB .

(a) Show that $p^2 - 32p + 192 = 0$.

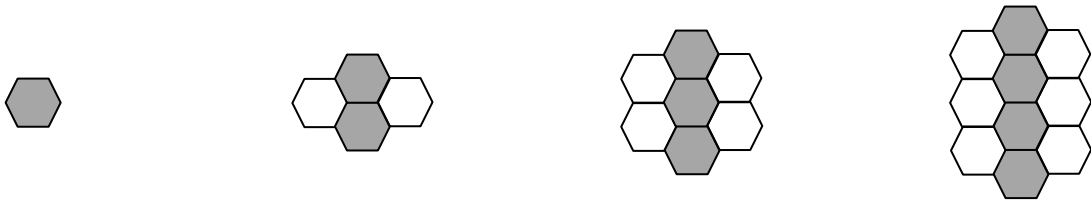
Answer

[2]

(b) Solve $p^2 - 32p + 192 = 0$ to find the length of AB .

Answer $AB = \dots\dots\dots$ cm [3]

15



Pattern 1

Pattern 2

Pattern 3

Pattern 4

(a) Complete the table.

Pattern number	1	2	3	4	5
Grey tiles	1	2	3	4	
White tiles	0	2	4	6	
Total	1	4	7	10	

[1]

(b) Find an expression, in terms of n , for

(i) the number of white tiles in Pattern n ,

Answer [1]

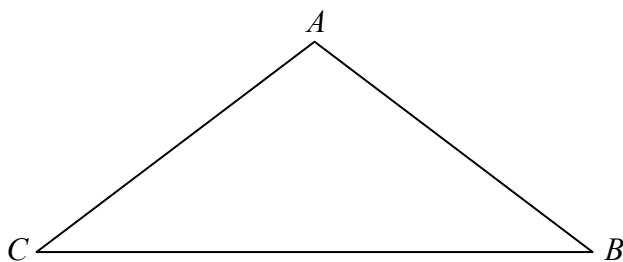
(ii) the total number of tiles in Pattern n .

Answer [1]

(c) Find the pattern number that has 133 tiles.

Answer Pattern [1]

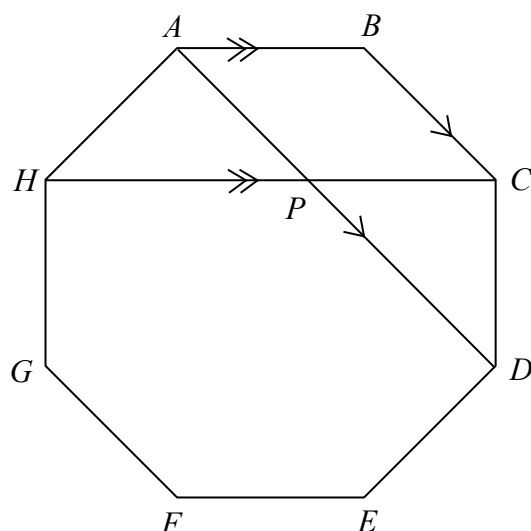
- 16 The diagram shows an isosceles triangle ABC .



- (a) Construct the bisector of angle ACB . [1]
- (b) Construct the perpendicular bisector of BC . [1]
- (c) The point D is such that $ABDC$ is a kite and $CD = 7$ cm.

On the diagram, construct kite $ABDC$. [1]

17



The diagram shows a regular octagon $ABCDEFGH$.

AB is parallel to HC .

BC is parallel to PD .

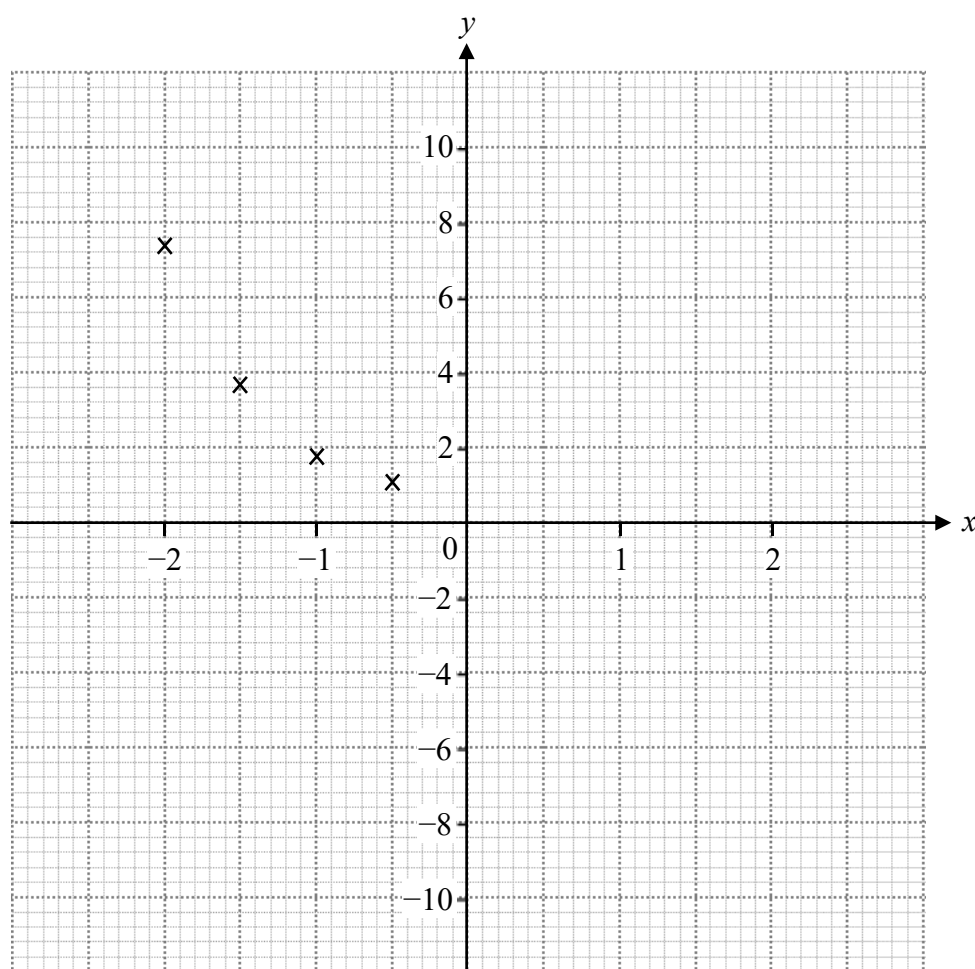
Find angle CPD .

Answer angle $CPD = \dots\dots\dots^\circ$ [2]

- 18 (a) Complete the graph of $y = 1 - 0.8x^3$ from $x = -2$ to $x = 2$.

Some points have been plotted for you.

Answer



[4]

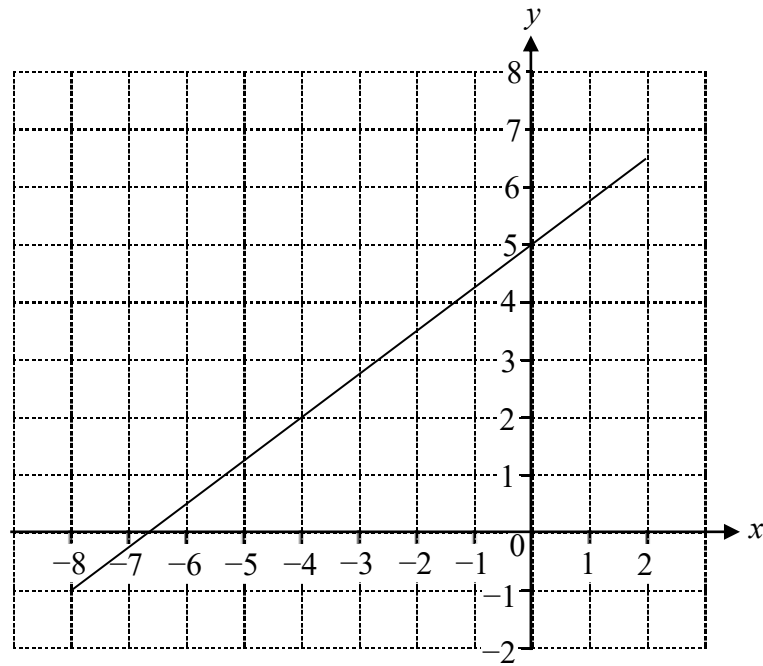
- (b) Using the same axes, draw the graph of $y = 3x + 6$.

[2]

- (c) Find the coordinates of the point where the two graphs intersect.

Answer (.....,) [1]

19



The points $A(-8, -1)$ and $B(-4, 2)$ lie on the line drawn on the grid above.

- (a) Calculate the length of AB .

Answer units [2]

- (b) Find the equation of the line drawn on the grid.

Answer [2]

- (c) $ABCD$ is a parallelogram, with AB and BC being two of its sides.

Given that C is the point $(1, 2)$, find D .

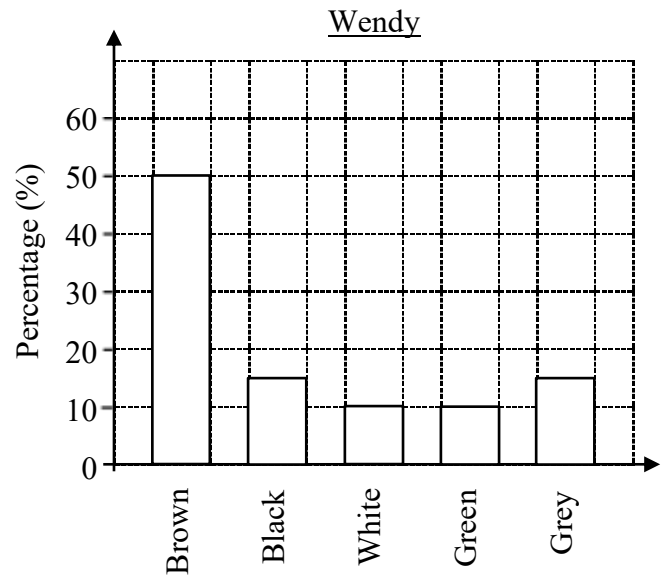
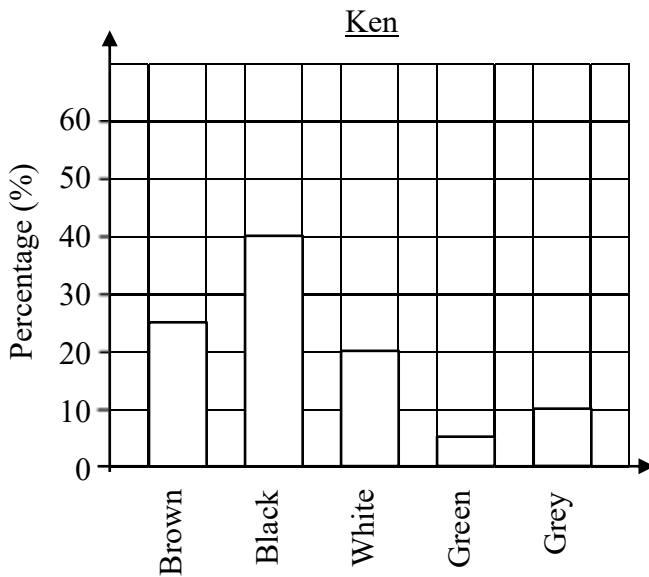
Answer $(\dots\dots\dots, \dots\dots\dots)$ [1]

- (d) Calculate the area of parallelogram $ABCD$.

Answer units² [1]

20 Ken and Wendy own a bag shop each.

The colours of bags Ken and Wendy sold in the month of June are summarised on the bar charts below.



(a) Ken says that the bar charts show that he sold more white bags than Wendy in June.

Explain why he might not be correct.

Answer

.....

.....[1]

(b) Wendy sold 42 black bags in June.

How many bags did Wendy sell in June?

Answer [1]

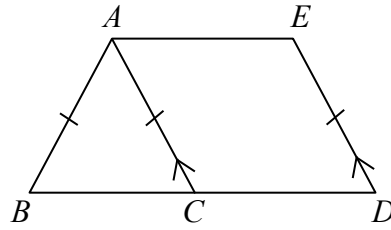
(c) Explain an advantage of representing these data as pie charts.

Answer

.....

.....[1]

21 (a)



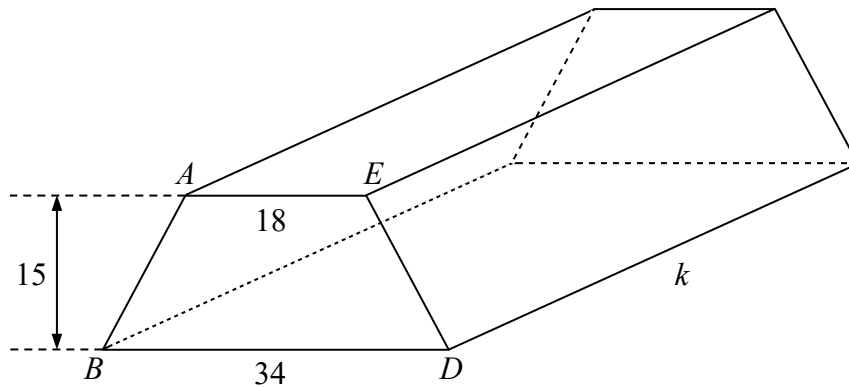
$ABDE$ is a quadrilateral with $AB = AC = DE$.

Given that AC is parallel to DE , explain why $ABDE$ is a trapezium.

Answer

[2]

(b)



The trapezium $ABDE$, in part (a), is the cross-section of this prism.

$ABDE$ has a height of 15 cm.

$AE = 18$ cm and $BD = 34$ cm.

The length of the prism is k centimetres.

Given that the volume of the prism is $19\,500\text{ cm}^3$, calculate k .

Answer $k = \dots\dots\dots$ [3]

END OF PAPER

